

Review of Likelihoods

STA 310: Homework 3

```
library(knitr)
library(magrittr)
library(kableExtra)
```

Exercise 1 and 2

Can be found in question 1 and 2 images file.

Exercise 3

Verify using a grid-search that your solution matches to the calculus based one, where you may assume for simplicity that $\sum_i x_i = 500$. You may assume 100 observations. (Hint: show that the approximated MLE is 5.)

```
# initial value .1 bc 0 is undefined
lambda <- seq(0.1, 10, by = 0.001)
# poisson log likelihood
loglik <- -100*lambda + 500*log(lambda)
# find the max value
lambda[which.max(loglik)]
```

```
## [1] 5
```

Exercise 4 (Derived from Chapter 2 of BMLR).

The hot hand in basketball. @Gilovich1985 wrote a controversial but compelling article claiming that there is no such thing as “the hot hand” in basketball. That is, there is no empirical evidence that shooters have stretches where they are more likely to make consecutive shots, and basketball shots are essentially independent events. One of the many ways they tested for evidence of a “hot hand” was to record sequences of shots for players under game conditions and determine if players are more likely to make shots after made baskets than after misses. For instance, assume we recorded data from one player’s first 5 three-point attempts over a 5-game period. We can assume games are independent, but we’ll consider two models for shots within a game:

- No Hot Hand (1 parameter): p_B = probability of making a basket (thus $1 - p_B$ = probability of not making a basket).
 - Hot Hand (2 parameters): p_B = probability of making a basket after a miss (or the first shot of a game); $p_{B|B}$ = probability of making a basket after making the previous shot.
- a. Fill out Table @ref(tab:hothandchp2)—write out the contribution of each game to the likelihood for both models along with the total likelihood for each model.
 - b. Given that, for the No Hot Hand model, $\text{Lik}(p_B) = p_B^{10}(1 - p_B)^{15}$ for the 5 games where we collected data, how do we know that 0.40 (the maximum likelihood estimator (MLE) of p_B) is a better estimate than, say, 0.30?

We know 0.40 is a better estimate than 0.30 because it maximizes the likelihood of observing the data. Plugging in the values, $\text{Lik}(.40) = .4^{10} * .6^{15}$ is larger than $\text{Lik}(.3) = .3^{10} * .7^{15}$. Therefore, .4 makes the observed sequence of makes and misses more probable, which is exactly what the MLE chooses.

- c. Find the MLEs for the parameters in each model, and then use those MLEs to determine if there's significant evidence that the hot hand exists using a likelihood ratio test (LRT). Be sure to specify the test and provide all details of your approach, including reproducible code used.

```
# no Hot Hand log-likelihood
logL0 <- function(pB) {
  10*log(pB) + 15*log(1 - pB)
}

# hot Hand log-likelihood
logL1 <- function(pB, pBB) {
  6*log(pB) + 9*log(1 - pB) +
  4*log(pBB) + 6*log(1 - pBB)
}

# no Hot Hand MLE
pB_hat0 <- 10/25

# hot Hand MLE
pB_hat1 <- 6/15
pBB_hat1 <- 4/10

l10 <- logL0(pB_hat0)
l11 <- logL1(pB_hat1, pBB_hat1)

#likelihood ratio test statistic

LR_stat <- -2 * (l10 - l11)

#chi squared pval

p_value <- pchisq(LR_stat, df=1, lower.tail=FALSE)

#tidy
list(
  MLE_no_hot_hand = pB_hat0,
  MLE_hot_hand = c(p_B = pB_hat1, p_B_after_B = pBB_hat1),
  logLik_no_hot_hand = l10,
  logLik_hot_hand = l11,
  LR_statistic = LR_stat,
  p_value = p_value
)

## $MLE_no_hot_hand
## [1] 0.4
##
## $MLE_hot_hand
##      p_B p_B_after_B
##      0.4      0.4
##
## $logLik_no_hot_hand
```

```
## [1] -16.82529
##
## $logLik_hot_hand
## [1] -16.82529
##
## $LR_statistic
## [1] 7.105427e-15
##
## $p_value
## [1] 0.9999999
```

The MLEs show that both models estimate the same probabilities ($p_b = .4$) and produce similar log-likelihoods. Consequently, the likelihood ratio test statistic is essentially zero with a p-value near 1, indicating no evidence of a hot hand in this data.

Game	First five shots	Likelihood (no hot hand)	Likelihood (hot hand)
1	BMMBB	$(1 - p_b)p_b^2(1 - p_b)^2$	$(1 - p_b) \cdot p_b \cdot p_b$
2	MBMBM	$(p_b^3)(1 - p_b)^2$	$p_b \cdot 1 - p_b$
3	MMBBB	$p_b^2(1 - p_b)^3$	$p_b \cdot p_b$
4	BMMMM	$(1 - p_b)p_b^3(1 - p_b)$	$(1 - p_b) \cdot p_b \cdot p_b$
5	MMMMM	p_b^5	$p_b \cdot p_b$